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Portfolio Optimization using Accelerated Quantum Simulation

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***Abstract*—Identifying the most suitable stocks for Portfolio stands as a critical challenge in the financial sector especially for a larger pool of assets. Quantum algorithms have emerged as a promising solution to the inherent challenges and offering the potential for exponential speedups in solving complex optimization problems. These algorithms are particularly suited for navigating the vast solution spaces associated with large asset pools, thus enabling more effective diversification strategies. Within this context, we explore the application of CUDA Quantum, an advanced quantum simulation framework and a quantum ready platform. Our study demonstrates that when applied to a set of 15 assets and more, CUDA Quantum significantly outperforms traditional CPU-based simulations in terms of computational speed and increases as the number of assets increase, highlighting the advantages of GPU acceleration in processing complex quantum algorithms. By extending our implementation to larger asset pools upto 30 assets, we provide empirical evidence of the substantial performance gains achievable with CUDA Quantum. Our findings illustrate the transformative potential of integrating quantum algorithms and GPU computing in addressing the challenge of portfolio optimization.**

***Index Terms*—Portfolio Optimisation, Quantum Computing, Quantum Simulations, Accelerated Systems, NVIDIA A100 GPUs, CUDA Quantum**

I. Introduction

In recent years, the intersection of quantum computing and financial modeling has ushered in a new era of portfolio optimization strategies. Traditional methods, while effective, often grapple with the computational complexities inherent in optimizing larger and diverse investment portfolios. Quantum computing, with its unparalleled processing capabilities, presents a groundbreaking solution to these challenges. Among the pioneering works in this domain is the study by Wang et al. (2024), which introduces a variational quantum eigensolver with a linear depth problem-inspired ansatz specifically designed for financial portfolio optimization based on mean-variance analysis. This method leverages the unique strengths of quantum computing to address optimization problems more efficiently than classical approaches .

Moreover, the empirical analysis conducted by Huot et al. (2024) on the Quantum Approximate Optimization Algorithm (QAOA) for knapsack-based financial portfolio optimization further illustrates the potential of quantum algorithms in enhancing the portfolio construction process. Their findings highlight the adaptability of quantum computing techniques in navigating the intricacies of market data to formulate optimized investment strategies .

The realm of quantum computing also extends its benefits to transfer learning and combinatorial optimization, as demonstrated by Montanez-Barrera et al. (2024). Their work showcases the applicability of quantum computing models in solving diverse financial problems, including portfolio optimization, by transferring optimal QAOA parameters across different computational models .

Access to real quantum hardware, especially those with a higher number of qubits and deeper gate configurations without significant noise, remains a challenging endeavor. Quantum simulators emerge as a practical solution, simulating quantum information processing on classical hardware and playing a pivotal role in the advancement of quantum computing. NVIDIA's cuQuantum, a part of the CUDA Quantum framework, exemplifies this innovation, offering an efficient platform for simulating larger quantum systems (M. Modani et.al 2022) . This framework significantly enhances the ability to explore complex quantum algorithms, including those pertinent to financial sectors like portfolio optimization. In our study, we harness the capabilities of the CUDA Quantum framework alongside cuQuantum's simulation prowess to develop a Portfolio Optimization Algorithm. Our analysis focuses on the algorithm's efficiency and the computational time required for optimization tasks, illustrating the practical applications and potential benefits of quantum simulations in financial modeling. This approach leverages recent advancements in GPU-accelerated quantum simulation, demonstrating the scalability and performance enhancements achievable through cuQuantum (Chen et al., 2024; Sugisaki et al., 2024).

We have implemented portfolio optimisation using cuda Quantum framework. In present work upto 30 assets were simulated.

II. Background

A. Mean Varianve Portfolio Opimisation

Mean Variance Portfolio Optimization (MVPO), established by Harry Markowitz in his seminal 1952 "Portfolio Selection" paper and further developed in his 1959 book, underpins portfolio optimisation theory by suggesting the achievement of an optimal portfolio through risk-return optimization, known as the efficient frontier. This concept has profoundly influenced financial management, guiding investors towards maximizing returns for a given risk level. However, the practical implementation of MVPO, complicated by the difficulty in accurately

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predicting future returns and covariances, has prompted advancements like the Black-Litterman model to refine portfolio allocations by incorporating investor insights with market equilibrium returns (Financial Analyst Journal, 1992).

The evolution of computational finance has introduced sophisticated techniques such as Monte Carlo simulations and machine learning to address MVPO's complexities. Remarkably, recent developments have seen the MVPO problem being reformulated as a Quadratic Unconstrained Binary Optimization (QUBO) problem, a move that significantly benefits from quantum computing's capabilities. This transformation enables the efficient exploration of solution spaces for portfolio optimization, marking a significant leap forward by employing quantum algorithms to overcome traditional computational limits and ushering in a new era of financial strategy optimization with enhanced efficiency and scalability.

B. Quadratic Unconstrained Binary Optimisation(QUBO)

The Quadratic Unconstrained Binary Optimization (QUBO) framework is a powerful tool for modeling various optimization problems encountered in finance, logistics, machine learning, and more. QUBO problems involve finding the optimal configuration of a set of binary variables to minimize a quadratic objective function. Due to their combinatorial nature, QUBO problems are particularly challenging for classical computing resources, especially as the problem size scales (Lucas, 2014).

Quantum computing, with its inherent parallelism, offers a promising solution to these challenges. Quantum algorithms, such as Quantum Annealing and the Quantum Approximate Optimization Algorithm (QAOA), have been specifically developed to tackle optimization problems modeled by QUBO formulations. These algorithms leverage quantum superposition and entanglement to explore multiple potential solutions in parallel, efficiently navigating the solution landscape to identify optimal or near-optimal solutions with significantly reduced computational effort (Farhi et al., 2014).

C. Vartional Quantum Eigensolver (VQE)

The Variational Quantum Eigensolver (VQE) stands as a hybrid quantum-classical algorithm designed to address one of the most challenging problems in quantum physics and chemistry—determining the ground state energy of a system. Introduced by Peruzzo et al. (2014), VQE utilizes the principles of variational methods in quantum mechanics, leveraging the power of quantum computers to handle the problem's quantum part while employing classical computers for optimization tasks. This collaborative approach allows VQE to efficiently approximate the ground state of a Hamiltonian, making it particularly suitable for early-stage quantum computers, which are limited by quantum noise and decoherence (Peruzzo et al., 2014; McClean et al., 2016).

Beyond its initial applications in quantum chemistry, VQE's methodology has been extended to tackle various optimization problems, illustrating its versatility and potential in fields such as finance, logistics, and machine learning (Orús et al., 2019). By encoding optimization problems into the Hamiltonian of a quantum system, VQE seeks the solution in the system's ground state, offering a novel approach to solving complex problems that are intractable for classical computing methods. This adaptability, coupled with the ongoing advancements in quantum computing technology, positions VQE as a pivotal tool in the burgeoning field of quantum optimization, promising significant impacts across multiple industries (Kandala et al., 2017; Orús et al., 2019).

D. Quantum Apporximate Optimisation Algorithm(QAOA)

The Quantum Approximate Optimization Algorithm (QAOA) has emerged as a promising quantum algorithm for addressing combinatorial optimization problems within and beyond the realm of classical computational capabilities.

Portfolio optimization, a critical task in financial management, involves selecting the best asset mix to maximize returns while minimizing risk. Classical methods often struggle with the combinatorial explosion as the number of assets increases. QAOA, by leveraging quantum computing's parallelism, offers a novel approach to efficiently approximate optimal portfolios, potentially outperforming classical optimization Buonaiuto, G et al. (2023), have begun to explore QAOA's utility in financial modeling, demonstrating its potential to revolutionize financial optimization tasks by harnessing quantum computational advantages.

E. Quantum Simulators and cuda Quantum

Quantum simulators represent an invaluable bridge between theoretical quantum computing and its practical applications, enabling researchers to model and analyze quantum phenomena on classical systems. As the quantum computing field navigates the nascent, noise-prone era of Noisy Intermediate-Scale Quantum (NISQ) devices, the need for quantum simulators becomes ever more critical. These simulators allow for the detailed study of quantum algorithms, system dynamics, and error correction techniques without the constraints of current quantum hardware limitations. Particularly, statevector simulation offers a direct method for visualizing quantum state evolutions through computational processes, albeit with the challenge of scaling exponentially with the addition of qubits, which demands significant computational resources for larger systems (Preskill, 2018).

Amidst various quantum simulation tools, NVIDIA's CUDA Quantum ecosystem, and specifically the cuQuantum SDK, stands out by leveraging the parallel processing prowess of GPUs to significantly enhance the efficiency and scalability of quantum simulations. This approach addresses the inherent limitations of classical simulation techniques, enabling more complex and larger-scale quantum simulations than previously possible. The cuQuantum SDK provides a suite of optimized tools and libraries designed for quantum computing, which, when combined with the computational power of NVIDIA GPUs, offers a robust platform for exploring quantum algorithms and systems with unprecedented detail and speed. This level of performance and scalability underpins the growing

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preference for CUDA Quantum and cuQuantum in quantum research and development, highlighting their potential to accelerate the advent of practical quantum computing applications (NVIDIA Corporation, 2021).

III. Methodology

Breif about the problem and its conversion

A. Dataset

The foundation of portfolio optimization lies in selecting a diverse set of assets to minimize risk and maximize return. For this study, we chose tickers representing a broad spectrum of industries, including technology, finance, consumer goods, and energy, to ensure diversification. The tickers include major corporations like AAPL, NKE, CO, and others, totaling thirty for a comprehensive analysis.

We utilized the adjusted closing price for 2 years from the Yahoo Finance API dated $2020/12/25$ to $2022/12/25$, which accounts for factors like dividends and stock splits, providing a more accurate representation of an asset's value. The choice of adjusted closing price ensures the analysis reflects the true return on investment.

After acquiring the stock prices, the we calculate the logarithmic returns of the stocks, which is a common practice in financial analyses to normalize price data. Later compute the mean (average returns) and covariance matrix of these log returns. The mean returns (μ) give an estimate of the expected return of each stock, and the covariance matrix (Σ) quantifies the degree to which returns on two stocks move in tandem.

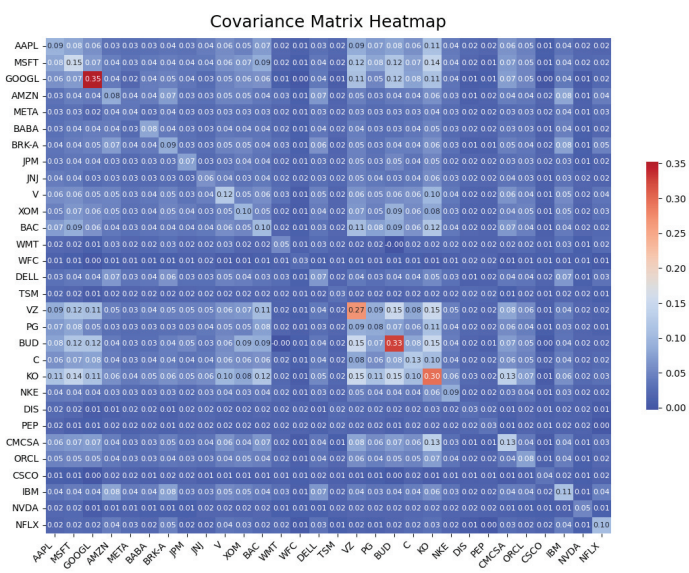

Fig. 1. The colored squares represent the correlation between two stocks. Colours like Yellow, Orange, Red indicate a positive correlation, meaning the stocks tend to move in the same direction. While colours Blue, Purple indicate a negative correlation, meaning the stocks tend to move in opposite directions. A gray color indicates a weak correlation, where there's no clear trend.

The covariance matrix as show in figure 1, represents the relationships between the returns of the selected assets, is visualized as a heatmap, illustrating the variance and correlation amongst the assets, foundational for portfolio optimization.

B. QUBO formation

The essence of QUBO is to reformulate the portfolio optimization problem, traditionally expressed in continuous terms, into a binary optimization problem suitable for quantum algorithms (Kochenberger, G et.al 2014).

In portfolio optimization, the goal is to select a combination of assets that minimizes risk while maximizing expected return, subject to budgetary and other constraints. The QUBO formulation of this problem involves defining a binary variable for each potential asset in the portfolio, where the variable indicates whether the asset is included (1) or excluded (0) from the portfolio. The objective function to be minimized (or maximized, depending on formulation) is then expressed as a quadratic function of these binary variables, incorporating terms for both the variance (risk) and expected return of the portfolio (Venturelli et.al 2019).

To formalize, let x_i denote the binary variable associated with the inclusion of asset i in the portfolio. The QUBO objective function can be expressed as:

$$H(x) = x^T Q x - \lambda \sum_{i=1}^{n} \mu_i x_i \tag{1}$$

where $H(x)$ is the Hamiltonian representing the portfolio optimization problem, Q is a matrix encoding the covariances between the assets, reflecting the portfolio risk, μ_i represents the expected return of asset i, λ is a regularization parameter balancing the trade-off between risk and return, x is the vector of binary variables x_i.

C. QUBO to Ising Model Conversion

The transformation from a QUBO problem to the Ising model is a pivotal step in utilizing quantum computing techniques for solving optimization problems. This conversion enables the application of quantum algorithms, such as the Quantum Approximate Optimization Algorithm (QAOA), to find solutions more efficiently than classical approaches. The QUBO framework, where the objective function is defined as

$$f(x) = x^T Q x \tag{2}$$

with x being a binary vector and Q a real symmetric matrix, is mapped to the Ising model by translating the binary variables into spin variables. The Ising model is expressed as

$$H(\sigma) = -\sum_{ij} J_{ij} \sigma_i \sigma_j - \sum_{i} h_i \sigma_i \tag{3}$$

where $H(\sigma)$ represents the Hamiltonian of the system, σ_i are the spin variables taking values in $\{-1, +1\}$, J_{ij} denotes the interaction between spins i and j, and h_i is an external magnetic field acting on spin i.

The correspondence between the binary variables x_i in QUBO and the spin variables σ_i in the Ising model is established through the transformation

$$\sigma_i = 2x_i - 1. \tag{4}$$

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This relation effectively translates the QUBO problem into an equivalent Ising model, where the quadratic terms in the QUBO objective are mapped to the interaction terms J_{ij} in the Ising Hamiltonian, and the linear terms correspond to the magnetic field h_i. Through this transformation, optimization problems can be encoded into the Hamiltonian of a quantum system and solved using quantum computing techniques.

D. Formulation of QAOA for Portfolio Optimization

In our study, we adopt the Quantum Approximate Optimization Algorithm (QAOA) to address the portfolio optimization problem, leveraging its potential to navigate the solution space more efficiently than classical algorithms. This section details the specific formulation of QAOA as derived from our implementation, focusing on the construction of the ansatz and the realization of the Hamiltonian components.

1) Problem Encoding and Hamiltonian Realization: The initial step in our QAOA formulation involves encoding the portfolio optimization problem into a quantum-compatible format. This encoding translates the classical problem of minimizing risk and maximizing returns into a quantum Hamiltonian, comprising both cost and mixer components.

a) Cost Hamiltonian: The cost Hamiltonian H_C is constructed to encapsulate the objective of minimizing portfolio variance while accounting for expected returns, reflecting the fundamental trade-off in portfolio optimization. Derived from the covariance matrix Σ and expected return vector μ, the Hamiltonian is expressed as:

$$H_C = \sum_{i,j} \sigma_{i,j} \Sigma_{i,j} - \lambda \sum_i \mu_i \sigma_i, \tag{5}$$

where $\sigma_{i,j}$ represent the spin variables corresponding to the inclusion or exclusion of assets in the portfolio, and λ is a regularization parameter that balances risk and return.

b) Mixer Hamiltonian: To ensure thorough exploration of the solution space, we employ a mixer Hamiltonian H_M, typically constituted by Pauli-X operators applied to each qubit. This component facilitates the transition between different portfolio configurations, aiding in the search for the global minimum:

$$H_M = \sum_i X_i. \tag{6}$$

2) QAOA Ansatz Implementation: Our implementation of the QAOA ansatz involves alternating applications of the cost and mixer Hamiltonians, controlled by sets of parameters $\{\gamma_k\}$ and $\{\beta_k\}$ respectively, for $k = 1, \ldots, p$, where p denotes the number of QAOA layers. The unitary operations associated with these Hamiltonians are given by:

$$U(H_C, \gamma) = e^{-i\gamma H_C}, \quad U(H_M, \beta) = e^{-i\beta H_M}, \tag{7}$$

with the full ansatz realized through the successive application of these unitaries for each layer of the algorithm.

3) Parameter Optimization and Solution Extraction: The core of the QAOA process in our approach involves optimizing the parameters $\{\gamma_k, \beta_k\}$ to minimize the expectation value of the cost Hamiltonian, which corresponds to identifying the optimal portfolio configuration. This optimization is performed using classical computational methods, iterating over the quantum circuit evaluation to converge towards the solution that offers the best balance between expected returns and associated risk.

E. Variational Quantum Eigensolver (VQE) in Portfolio Optimization

The Variational Quantum Eigensolver (VQE) plays a pivotal role in solving the eigenvalue problem associated with the Hamiltonian representing the portfolio optimization challenge. By leveraging the principles of quantum mechanics, VQE efficiently approximates the ground state energy of the Hamiltonian, which corresponds to the optimal portfolio allocation in our context.

1) VQE Framework: VQE is a hybrid quantum-classical algorithm that minimizes the expectation value of the Hamiltonian of a quantum system. It operates by iteratively adjusting the parameters of a parameterized quantum circuit (ansatz) to find the state that minimizes the energy of the system. The expectation value of the Hamiltonian, given a quantum state $|\psi(\theta)\rangle$ parameterized by θ, is calculated as:

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle, \tag{8}$$

where H is the Hamiltonian encoding the portfolio optimization problem. The goal of VQE is to find the set of parameters θ^* that minimizes $E(\theta)$, which is indicative of the ground state energy of H.

2) Role of Classical Optimizer: The classical optimizer in the VQE algorithm is responsible for updating the parameters θ of the quantum circuit based on the feedback received from the quantum computations. After each evaluation of the expectation value $E(\theta)$, the classical optimizer suggests new parameters in an attempt to find the minimum energy configuration. Various optimization techniques can be employed, including gradient descent, Nelder-Mead, or COBYLA, each with its advantages in different problem landscapes.

3) QAOA Circuit Evolution and VQE: In the context of portfolio optimization, the QAOA circuit serves as the ansatz for the VQE algorithm. The QAOA parameters $\{\gamma, \beta\}$ are analogous to the θ in the VQE formulation, controlling the application of cost and mixer Hamiltonians. As the circuit evolves through the iterative VQE process:

$$|\psi(\gamma, \beta)\rangle = U(H_M, \beta) U(H_C, \gamma) |\psi_0\rangle, \tag{9}$$

where $|\psi_0\rangle$ is the initial state, typically chosen as a superposition of all possible states to ensure broad coverage of the solution space.

The integration of VQE with QAOA enables the efficient approximation of the optimal portfolio by exploiting quantum superposition and entanglement to explore and evaluate a vast number of potential allocations simultaneously. Through

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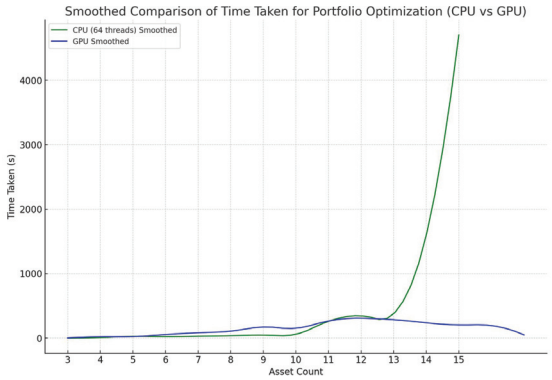

Fig. 2. Time comparison of CPU vs GPU till 15 assets

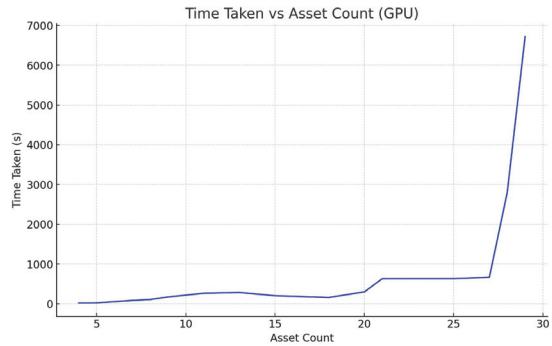

Fig. 3. Time taken for 30 assets simulation on GPU

the iterative refinement of γ and β, guided by the classical optimizer, the algorithm converges to a solution representing an optimal or near-optimal portfolio allocation.

IV. Result and Discussion

We implemented the Portfolio Optimisation using cuda Quantum, both gpu and cpu enabled. Study was conducted on NVIDIA A100 80GB Tensor Core GPU system with Dual AMD Rome 7742, 128 cores CPU, and system memory of 2 TB. For the classical optimiser for VQE, COBYLA optimiser was used, as present in the cuda Quantum platform.

The results depicted in Figures 2 and 3 illustrate the performance comparison between CPU and GPU simulations for Portfolio Optimization (PO) across different asset counts. Figure 2 presents a comparison of the time taken to complete Portfolio Optimisation tasks with both CPU (64 threads) and GPU implementations. Notably, the CPU time remains relatively stable and then exhibits a dramatic increase after asset count of 15. Conversely, the GPU time maintains a consistent advantage over the CPU, showing minimal increase as the asset count grows and only a moderate rise at the same threshold.

Figure 3 focuses exclusively on GPU performance, further examining the scalability of this approach. The GPU maintains a relatively flat curve of computational time up to an asset count of 27, beyond which we observe a sharp spike in time required. This inflection point suggests the onset of computational complexity that grows significantly as the number of assets increases, a characteristic often associated with combinatorial optimization problems in Portfolio Optimisation.

V. Conclusion and Future Scope

In conclusion, this study presents a comprehensive evaluation of the efficacy of GPU-accelerated quantum simulations in the realm of Portfolio Optimization, with a focus on the dual aspects of accuracy and performance. Comparative analysis reveals that GPU implementations offer significant advantages over CPU in terms of speed and efficiency, confirming the utility of GPU technologies in handling complex quantum simulations. Performance consistency up to 27 qubits and the exponential increase in execution time beyond this point highlight the scalability challenges inherent in quantum computing. These findings underscore the need for continued advancements in quantum simulation technologies to harness the full potential of quantum computing in financial modeling and extend its applicability to larger asset pools, thereby achieving more sophisticated investment strategies.

The current implementation of the algorithm, while demonstrating the substantial benefits of GPU acceleration in Portfolio Optimization, must be rigorously tested across diverse scenarios to ascertain its robustness and versatility. A critical area of future research involves analyzing performance on systems equipped with multiple GPUs and extending this to multi-node architectures to evaluate scalability in a distributed computing environment. These investigations will help in identifying the optimal system configurations required to handle the computational demands of larger asset pools efficiently.

Further analysis is also planned using TensorNet, which could offer insights into the benefits and potential trade-offs of alternative quantum simulation frameworks. Anomalies observed in the execution time, particularly for certain asset counts, prompt the need for a deeper exploration to understand the underlying causes and implications. Such irregularities warrant a comprehensive analysis to ensure the reliability and predictability of performance across all possible investment scenarios.

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